THEA – THE LIGHT MATRIX

Full LaTeX Tower v1.3.2

exact closure, golden selection, chirality, spectra, invariant theory, numerical walls, and the LUCA handoff

Independent audit edition
Buenos Aires + Ancient Korinthos lineage
Prepared 9 August 2026

Abstract

This document is a source-grounded, independently recomputed consolidation of the seven attached source artifacts listed in Chapter 1. It is a new versioned artifact: Light Matrix v1.3.1 and LUCA v0.1/v0.2 remain frozen.

The exact mathematical core survives the audit: Euler forces twelve pentagons; the hexagonal closure norm is multiplicative; the lifted integer recursion has spectrum $\{\varphi^2, 1, -1, \varphi^{-2}\}$; the golden selector produces independently closed Goldberg shells; the golden ray has the exact bearing $\arctan(\sqrt{15} - 2\sqrt{3})$; the complete C_{60} adjacency characteristic polynomial and Fiedler radical reproduce exactly; the Klein syzygy closes symbolically; and the finite browser graph calculations reproduce the reported low-mode structure.

The audit also changes several labels. The proposed constant $2\pi/(5\sqrt{3})$ is an exact consequence of a continuum matching argument, but the convergence of a chosen discrete graph family to that constant is not proved by that argument. The A_2 root lattice and the $SU(3)$ weight lattice are not literally the same lattice; they are dual, commensurable hexagonal lattices with index three between them. Random basin counting is a finite numerical experiment, not a theorem. The float64 comparison in v1.3.1 used a rounded version of the exact integer on its reference side; the corrected comparison still finds forward stability, but at a true error floor near 10^{-15}. Finally, the live LUCA v0.2 code includes 33 points in its leaf-count regression, whereas the handoff scroll's published coefficients correspond to 32 points after excluding the initial constant e.

The governing rule is therefore retained in its strongest form: target is not result; theorem, numerical trend, design mapping, physical hypothesis, and metaphor never share a label merely because they share a glyph.

The pentagons hold. The hexes pay. The matrix remembers which is which.

$$P = 12, \quad \chi = 2, \quad \mathrm{spec}(\mathcal{M}_{\mathrm{light}}) = \{\varphi^2, 1, -1, \varphi^{-2}\}.$$

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The principal tabular receipts are intentionally embedded in the mathematical narrative rather than separated from their hypotheses. Their locations are:

Source inventory
: Chapter 1: the seven audited artifacts and their roles.

Golden shell catalogue
: Chapter 4: (k_n, ℓ_n), T_n, cage size, hexagon count, and chirality angle.

Light-matrix eigenbasis
: Chapter 5: all four modes and their exact eigenvectors.

C_{60} **polynomial**
: Chapter 6: the complete adjacency characteristic factorization.

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: Chapter 7: the v1.3.1 Goldberg-builder/Lanczos output through C_{17660}.

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: Chapter 10: significand budgets and exact-integer scale limits.

LUCA comparisons
: Chapter 11: finite-N angle sweep and the 32/33-point leaf-regression distinction.

Independent audit
: Chapter 14: all 38 checks, statuses, verdicts, and compact results; full values remain in the JSON receipt.

Chapter 1

The Contract: Status Before Symbol

1.1 Source inventory and lineage

Artifact	file	Use in this tower
THEA core	`Thea.md`	Exact topology, closure algebra, Fibonacci selector, light matrix, spectral receipts, and the honest physical boundary.
Light Matrix shell	`shell__thea_light_matrix_v1.3.1.html`	Nineteen interactive sections, exact C_{60} kernel, Goldberg builder, spectral routines, $A_2/SU(3)$ lane, Klein lane, and numerical-architecture lane.
LUCA v0.1	`shell__eml_luca_spiral_v0_1.html`	EML operator, reduction tree, branch receipts, and target/current/error verification.
LUCA v0.2	`shell__eml_luca_spiral_v0_2.html`	Live Faddeev–LeVerrier factorization, golden selector readout, golden-ray angle, finite-N Vogel sweep, and leaf-count null result.
Angle handoff	`THE_ANGLE_OF_THE_GOLDEN_RAY.md`	Radical angle, asymptotic alternating convergence, anti-palindromic characteristic polynomial, and build contract.
Twelve Paths	`THE_12_PATHS_OF_THE_FRACTAL_MAGE.md`	Proof by kernel, target $\neq$ result, immutable versions, and open receipts.
Kernel grimoire	`KERNELIMAGIC.md`	False convergence, pre-allocation cost gates, deterministic certificates, and portable paths.

1.2 The status grammar

Every displayed claim in v1.3.2 belongs to one of the following classes.

EXACT Follows from algebra, topology, integer arithmetic, or a symbolic identity shown in the document.

COMPUTED Reproduced by a named finite algorithm at a stated precision, sample count, graph depth, or iteration count.

DESIGN CHOICE A visualization, parameterization, tolerance, mapping, sampling schedule, or user-interface rule.

HYPOTHESIS A proposed physical interpretation that still requires discriminating evidence.

EXTERNAL / METAPHOR A statement inherited from another artifact or source and not independently proved here.

CORRECTION A source statement whose algebra, label, indexing, or implementation receipt changes in v1.3.2.

v1.3.2 correction

The HTML header advertises additions through Section XVII, yet the baked document contains a Section XVIII; the HUD count of 19 includes the birth section plus XVIII. v1.3.2 makes the section inventory explicit. It also repairs the abstract glyph C_{00} to C_{60} and replaces "zero dependencies" by the narrower, accurate statement that the mathematical kernel is dependency-free while presentation may use KaTeX.

1.3 Audit outcome in one page

The independent Python/SymPy/NetworkX/NumPy audit executed 38 checks and returned 38 passes. "Pass" here means that either the source statement was reproduced or its correction was independently established. The most important changes are:

1. The sequence T_{n+1}/T_n does not monotonically climb to φ^2; it alternates around the limit.
2. Golden-selected shells form a catalogue of independently closed shells. They are not an exactly nested fixed Goldberg transform with linear multiplier φ.
3. Section VIII applies Lanczos to a graph Laplacian, not to the 4×4 light matrix.
4. The continuum coefficient $2\pi/(5\sqrt{3})$ is exact conditional algebra. The graph-limit arrow remains a conjectural asymptotic identification supported by finite computation.
5. $Q(A_2)$ and $P(A_2)$ are not literally equal. The common hexagonal quadratic form must be stated with normalization.
6. The Klein syzygy is exact; basin counts are finite, seed-dependent numerical coverage experiments.
7. Binary256 has 237 bits of significand precision, not a "256-bit mantissa."
8. The v1.3.1 recurrence comparator shared the same binary64 rounding in both operands. A rational comparison against the exact integer gives a true maximum relative error of about 1.02×10^{-15} over levels 0–159.
9. LUCA branch identities require domains and branch sheets. A sample lock is a sieve, not a proof.
10. The live LUCA v0.2 regression uses 33 spiral nodes; the 32-point handoff values are recovered only after excluding e.

Chapter 2

Closed Topology: Why Twelve Cannot Move

2.1 General trivalent curvature ledger

Let f_p be the number of p-gonal faces in a connected, closed, trivalent planar graph embedded on S^2. Then

$$3V = 2E, \qquad \sum_{p\geq 3} p f_p = 2E, \qquad V - E + \sum_{p\geq 3} f_p = 2. \tag{2.1}$$

Eliminating V and E gives the exact combinatorial curvature identity

$$\boxed{\sum_{p\geq 3}(6-p)f_p = 12.} \tag{2.2}$$

This is more general than the fullerene statement. Triangles contribute $+3$, squares $+2$, pentagons $+1$, hexagons zero, and larger faces negative charge.

2.2 The fullerene specialization

For a pentagon/hexagon fullerene, $f_5 = P$, $f_6 = H$, and every other f_p vanishes. Equation (2.2) becomes

$$(6-5)P + (6-6)H = 12, \tag{2.3}$$

so

Exact statement

$$\boxed{P = 12.} \tag{2.4}$$

The assumptions are part of the theorem: connected, closed, spherical, trivalent, and containing only pentagonal and hexagonal faces.

The remaining counts follow:

$$\boxed{E = \frac{3V}{2}, \qquad H = \frac{V}{2} - 10, \qquad F = P + H = \frac{V}{2} + 2.} \tag{2.5}$$

For C_{60},

$$(V, E, F, P, H, \chi) = (60, 90, 32, 12, 20, 2). \tag{2.6}$$

2.3 Discrete Gauss–Bonnet

Assign the face charge

$$K_p = \frac{\pi}{3}(6 - p). \tag{2.7}$$

Then

$$\sum_f K_f = \frac{\pi}{3} \sum_p (6 - p) f_p = \frac{\pi}{3} \cdot 12 = 4\pi. \tag{2.8}$$

Honest boundary

The equality is exact combinatorial topology. It does not assert that a molecular fullerene carries uniform smooth Gaussian curvature, nor that graph curvature is physical spacetime curvature. It is a ledger of angular defect for this trivalent tiling class.

Chapter 3

The Hexagonal Closure Algebra

3.1 The sixth-root coordinate

Set

$$\zeta_6 = e^{i\pi/3} = \frac{1}{2} + i\frac{\sqrt{3}}{2}, \qquad \zeta_6^2 = \zeta_6 - 1. \tag{3.1}$$

A lattice displacement is

$$z = k + \ell\zeta_6, \qquad k, \ell \in \mathbb{Z}. \tag{3.2}$$

Its squared Euclidean length is

$$\boxed{T = N(z) = z\bar{z} = k^2 + k\ell + \ell^2.} \tag{3.3}$$

This is the alternate sixth-root parameterization of the Eisenstein lattice. If one instead uses the conventional cube root $\omega = e^{2\pi i/3}$, the same lattice is written with the norm $a^2 - ab + b^2$. The sign difference is a basis convention, not a contradiction.

3.2 Multiplication as an integer matrix

Multiplication by $z = k + \ell\zeta_6$ sends $a + b\zeta_6$ to

$$(k + \ell\zeta_6)(a + b\zeta_6) = (ka - \ell b) + (\ell a + (k + \ell)b)\zeta_6. \tag{3.4}$$

Therefore

$$\begin{pmatrix} a' \\ b' \end{pmatrix} = M_{k,\ell} \begin{pmatrix} a \\ b \end{pmatrix}, \qquad \boxed{M_{k,\ell} = \begin{pmatrix} k & -\ell \\ \ell & k + \ell \end{pmatrix}.} \tag{3.5}$$

The determinant is the norm:

$$\det M_{k,\ell} = k(k + \ell) + \ell^2 = T. \tag{3.6}$$

With the hexagonal metric

$$Q = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix}, \tag{3.7}$$

direct multiplication gives

$$\boxed{M_{k,\ell}^{\mathsf{T}} Q M_{k,\ell} = TQ.} \tag{3.8}$$

Thus $M_{k,\ell}/\sqrt{T}$ is Q-orthogonal: an exact scale-plus-rotation in the lattice metric.

3.3 Angle and chirality coordinate

The exact lattice bearing is

$$\boxed{\theta_{k,\ell} = \arg(k + \ell\zeta_6) = \operatorname{atan2}(\sqrt{3}\,\ell, 2k + \ell).} \tag{3.9}$$

For the canonical wedge $k \geq \ell \geq 0$, $0 \leq \theta \leq \pi/6$. The boundary classes $\ell = 0$ and $k = \ell$ are achiral; strict interior pairs occur in mirror-related chiral classes (k, ℓ) and (ℓ, k).

3.4 Composition and multiplicativity

Let $g = a + b\zeta_6$. Then

$$(a + b\zeta_6)(k + \ell\zeta_6) = (ak - b\ell) + (a\ell + bk + b\ell)\zeta_6, \tag{3.10}$$

$$(k', \ell') = (ak - b\ell,\ a\ell + bk + b\ell), \tag{3.11}$$

$$M_{a,b}M_{k,\ell} = M_{k',\ell'}, \tag{3.12}$$

$$T' = (a^2 + ab + b^2)T. \tag{3.13}$$

This is the exact fixed-operator closure lane. The generator $(1, 1)$ has norm 3 and produces the nested leapfrog count tower

$$T_n = 3^n, \qquad V_n = 20\,3^n, \tag{3.14}$$

with $C_{20} \to C_{60} \to C_{180} \to C_{540} \to \cdots$ at the count level.

3.5 Selected is not nested

A fixed Goldberg operator has integer area multiplier

$$S = a^2 + ab + b^2 \in \mathbb{Z} \tag{3.15}$$

and linear multiplier $\sqrt{S}$. Exact golden nesting would require $S = \varphi^2$, impossible because S is an integer and φ^2 is irrational. Hence

Exact statement

$$\boxed{\text{No fixed exactly nested Goldberg transform has linear ratio } \varphi.} \tag{3.16}$$

The Fibonacci construction below is a sequence of independently closed shells whose successive radius ratios approach φ.

Chapter 4

The Golden Selector and the Ray with a Bearing

4.1 Fibonacci projective dynamics

Let

$$F_\varphi = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \begin{pmatrix} k_{n+1} \\ \ell_{n+1} \end{pmatrix} = F_\varphi \begin{pmatrix} k_n \\ \ell_n \end{pmatrix}, \qquad (k_0, \ell_0) = (1, 0). \tag{4.1}$$

Then

$$(k_n, \ell_n) = (F_{n+1}, F_n), \qquad \operatorname{spec}(F_\varphi) = \{\varphi, -\varphi^{-1}\}. \tag{4.2}$$

For $r_n = k_n/\ell_n$,

$$r_{n+1} = 1 + \frac{1}{r_n}, \qquad r_{n+1} - \varphi = -\frac{r_n - \varphi}{\varphi r_n}. \tag{4.3}$$

Therefore $r_n \to \varphi$, with alternating error and asymptotic contraction $-\varphi^{-2}$.

4.2 Triangulation sequence

Define

$$T_n = k_n^2 + k_n \ell_n + \ell_n^2. \tag{4.4}$$

The first values are

$$1, 3, 7, 19, 49, 129, 337, 883, 2311, 6051, \ldots \tag{4.5}$$

and

$$\boxed{T_{n+3} = 2T_{n+2} + 2T_{n+1} - T_n.} \tag{4.6}$$

The characteristic polynomial is

$$r^3 - 2r^2 - 2r + 1 = (r + 1)(r^2 - 3r + 1), \tag{4.7}$$

with roots $-1, \varphi^2, \varphi^{-2}$. Binet reduction gives

$$\boxed{T_n = \frac{2}{5}\left(\varphi^{2n+2} + \varphi^{-2n-2}\right) - \frac{1}{5}(-1)^n.} \tag{4.8}$$

Consequently

$$\frac{T_{n+1}}{T_n} \longrightarrow \varphi^2, \tag{4.9}$$

but the convergence is alternating rather than monotone.

v1.3.2 correction

The v1.3.1 sentence "the ratio T_{n+1}/T_n climbs to φ^2" is replaced by "the ratio alternates about and converges to φ^2." The computed sequence begins $7/3 = 2.3333$, $19/7 = 2.7143$, $49/19 = 2.5789$, $129/49 = 2.6327$.

4.3 Closed-shell catalogue

For each Fibonacci pair,

$$V_n = 20T_n, \quad E_n = 30T_n, \quad P_n = 12, \quad H_n = 10(T_n - 1), \quad F_n = 10T_n + 2. \tag{4.10}$$

n	(k_n, ℓ_n)	T_n	cage	H_n	θ_{k_n,ℓ_n}
0	(1,0)	1	C_{20}	0	0°
1	(1,1)	3	C_{60}	20	30°
2	(2,1)	7	C_{140}	60	19.106605°
3	(3,2)	19	C_{380}	180	23.413224°
4	(5,3)	49	C_{980}	480	21.786789°
5	(8,5)	129	C_{2580}	1280	22.411902°
6	(13,8)	337	C_{6740}	3360	22.172618°
7	(21,13)	883	C_{17660}	8820	22.264023°

4.4 The exact golden-ray angle

Taking $k/\ell \to \varphi$ in Equation (3.9),

$$\theta_\varphi = \arctan \frac{\sqrt{3}}{2\varphi + 1} = \arctan \frac{\sqrt{3}}{\sqrt{5} + 2} \tag{4.11}$$

$$= \boxed{\arctan(\sqrt{15} - 2\sqrt{3})} \tag{4.12}$$

$$= 0.388139515\ldots \ \text{rad} = 22.238756093\ldots^\circ . \tag{4.13}$$

The radical step uses $(\sqrt{5} + 2)(\sqrt{5} - 2) = 1$.

4.5 Why the angular error inherits the alternating mode

Let

$$\Theta(r) = \arctan \frac{\sqrt{3}}{2r + 1}. \tag{4.14}$$

Since Θ is differentiable and $\Theta'(\varphi) \neq 0$,

$$\theta_n - \theta_\varphi = \Theta'(\varphi)(r_n - \varphi) + \mathcal{O}((r_n - \varphi)^2). \tag{4.15}$$

Therefore

$$\boxed{\frac{\theta_{n+1} - \theta_\varphi}{\theta_n - \theta_\varphi} \longrightarrow -\varphi^{-2}.} \tag{4.16}$$

This is an asymptotic equality. At $n = 16$ the independent float64 recomputation gives

$$-0.3819660384, \qquad -\varphi^{-2} = -0.3819660113. \tag{4.17}$$

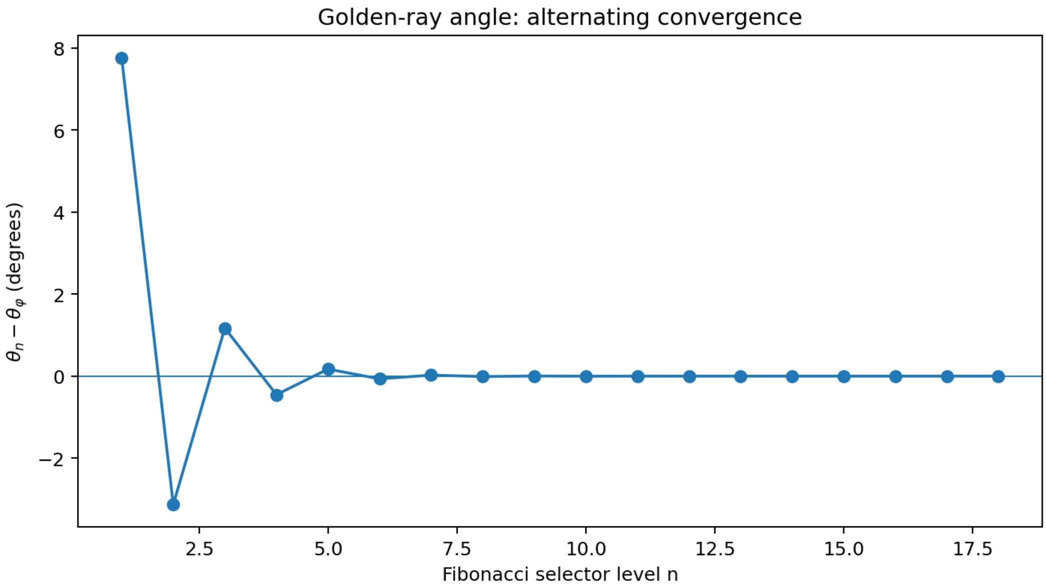

Figure 4.1: The golden-ray angular deviation alternates and contracts. The signed zigzag is the projective -1 mode made visible through the smooth angle map.

Chapter 5

The Exact Light Matrix

5.1 Symmetric-square lift

The pair recursion is

$$k' = k + \ell, \qquad \ell' = k. \tag{5.1}$$

Lift to quadratic monomials

$$u_n = \begin{pmatrix} k_n^2 \\ k_n \ell_n \\ \ell_n^2 \end{pmatrix}. \tag{5.2}$$

Then

$$\begin{pmatrix} (k+\ell)^2 \\ (k+\ell)k \\ k^2 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 2 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}}_{B} \begin{pmatrix} k^2 \\ k\ell \\ \ell^2 \end{pmatrix}. \tag{5.3}$$

Append the topological coordinate $P_n = 12$:

$$s_n = \begin{pmatrix} k_n^2 \\ k_n \ell_n \\ \ell_n^2 \\ P_n \end{pmatrix}, \qquad \boxed{s_{n+1} = \mathcal{M}_{\text{light}} s_n}, \tag{5.4}$$

where

$$\boxed{\mathcal{M}_{\text{light}} = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.} \tag{5.5}$$

5.2 Faddeev–LeVerrier certificate

Starting with $B_0 = I$ and

$$c_j = -\frac{1}{j}\operatorname{tr}(\mathcal{M}_{\text{light}} B_{j-1}), \qquad B_j = \mathcal{M}_{\text{light}} B_{j-1} + c_j I, \tag{5.6}$$

the coefficient list is

$$\boxed{(1, -3, 0, 3, -1)}. \tag{5.7}$$

Thus

$$p(\lambda) = \lambda^4 - 3\lambda^3 + 3\lambda - 1 \tag{5.8}$$
$$= (\lambda - 1)(\lambda + 1)(\lambda^2 - 3\lambda + 1). \tag{5.9}$$

The coefficients are anti-palindromic:

$$p(\lambda) = -\lambda^4 p(1/\lambda). \tag{5.10}$$

Therefore the spectrum is closed under reciprocal inversion.

5.3 Spectrum and eigenmodes

$$\boxed{\mathrm{spec}(\mathcal{M}_{\mathrm{light}}) = \{\varphi^2, 1, -1, \varphi^{-2}\}.} \tag{5.11}$$

A convenient eigenbasis is

λ	eigenvector	role
φ^2	$(\varphi^2, \varphi, 1, 0)^{\mathsf{T}}$	dominant quadratic area/atom growth
1	$(0, 0, 0, 1)^{\mathsf{T}}$	fixed $P = 12$ topological coordinate
-1	$(-2, 1, 2, 0)^{\mathsf{T}}$	alternating finite-size correction
φ^{-2}	$(\varphi^{-2}, -\varphi^{-1}, 1, 0)^{\mathsf{T}}$	contracting reciprocal mode

The eigenvalues are not fitted to shell data; they follow from the integer matrix. The design interpretation "grow, hold, overshoot, decay" is useful, but the matrix itself is the exact object.

5.4 The shell number as a linear observable

The triangulation number is

$$T_n = (1, 1, 1, 0)s_n. \tag{5.12}$$

Projecting the matrix recurrence onto this observable yields Equation (4.6). The P coordinate does not enter T_n; it survives as the independent eigenvalue-one mode.

Chapter 6

The Exact C_{60} Spectral Boundary

6.1 Independent graph certificate

The audit reconstructs the truncated icosahedron by truncating each directed incidence of the icosahedron. The resulting graph has

$$V = 60, \quad E = 90, \quad \deg(v) = 3, \quad f_5 = 12, \quad f_6 = 20, \quad \chi = 2. \tag{6.1}$$

These checks are integer and planar-embedding facts.

6.2 Complete adjacency characteristic polynomial

For adjacency matrix A_{60},

$$\chi_{A_{60}}(x) = (x-3)(x-1)^9(x+2)^4(x^2-x-3)^5(x^2+x-4)^4 \tag{6.2}$$

$$\times (x^2+x-1)^5(x^2+3x+1)^3 \tag{6.3}$$

$$\times (x^4-3x^3-2x^2+7x+1)^3. \tag{6.4}$$

The SymPy exact characteristic polynomial of the independently reconstructed graph factors identically.

6.3 The golden lower boundary

The least root comes from x^2+3x+1:

$$\boxed{\lambda_{\min}(A_{60}) = \frac{-3-\sqrt{5}}{2} = -\varphi^2,} \tag{6.5}$$

with multiplicity three.

6.4 Graph Laplacian and Fiedler radical

Because the graph is cubic,

$$L_{60} = 3I - A_{60}. \tag{6.6}$$

The first positive eigenvalue is

$$\boxed{\lambda_2(L_{60}) = \frac{9}{4} - \frac{\sqrt{2}}{8}\left(\sqrt{10} + 2\sqrt{19-\sqrt{5}}\right)} = 0.2434017461399\ldots \tag{6.7}$$

and is a root of

$$x^4 - 9x^3 + 25x^2 - 22x + 4 = 0 \tag{6.8}$$

with multiplicity three. The independent numerical residual is below 10^{-12}.

6.5 Low bands

The first nonzero Laplacian bands of C_{60} are

value	multiplicity	exact origin / interpretation
0.2434017461	3	Fiedler triplet, Equation (6.7)
0.6972243623	5	$(5-\sqrt{13})/2$
1.1797507493	3	quartic sibling
1.4384471872	4	$(7-\sqrt{17})/2$
2	9	adjacency eigenvalue 1
2.3819660113	5	$3-(\varphi^{-1})$ branch

The multiplicities $3, 5, 3+4$ are compatible with low spherical-harmonic dimensions under icosahedral splitting. Compatibility is not identity with a continuum field theory.

Honest boundary

An adjacency or graph-Laplacian eigenvalue is not automatically a molecular orbital energy. The exact graph spectrum is a valid Hückel-style combinatorial object; a physical Hamiltonian requires geometry, orbital integrals, interactions, and other chemistry.

Chapter 7

Graph Towers, Renormalization, and the Conditional Continuum Constant

7.1 Three families that must remain separate

Golden-selected catalogue $(k_n, \ell_n) = (F_{n+1}, F_n)$. Independently closed Goldberg shells with $T = 1, 3, 7, 19, \ldots$ and asymptotic radius ratio φ.

Fixed GC / leapfrog lane Repeated application of one integer generator, e.g. $(1, 1)$ with area multiplier 3. Exactly nested at the operator level, linear scale $\sqrt{3}$.

WELD lineage The source scroll reports a separate exact-ID chamfer family $C_{60} \to C_{240} \to C_{960} \to \cdots$ with multiplier 4.

No spectral value may be transferred from one family to another without recomputation.

7.2 Closure certificate

A candidate shell may be named closed only if

$$P = 12, \qquad \chi = 2, \qquad \partial E = 0, \qquad E_{\text{nonmanifold}} = 0, \qquad \deg(v) = 3 \ \forall v. \tag{7.1}$$

Formula counts are necessary but not sufficient for a welded indexed mesh.

7.3 Live v1.3.1 golden-shell receipt

The current HTML kernel was executed directly in Node, without reimplementing its Goldberg builder or Lanczos routine. The result is:

(k, ℓ)	T	V	λ_2	$T\lambda_2$	deviation from $2\pi/(5\sqrt{3})$
$(1, 0)$	1	20	0.7639320225	0.763932022500	+0.038412276807
$(1, 1)$	3	60	0.24340174614	0.730205238420	+0.004685492726
$(2, 1)$	7	140	0.10364786791	0.725535075371	+0.000015329677
$(3, 2)$	19	380	0.0381363955918	0.724591516244	-0.000928229449
$(5, 3)$	49	980	0.0147874462246	0.724584865005	-0.000934880689
$(8, 5)$	129	2580	0.0056176826499	0.724681061837	-0.000838683857
$(13, 8)$	337	6740	0.00215058792037	0.724748129166	-0.000771616528
$(21, 13)$	883	17660	0.000820819134337	0.724783295619	-0.000736450074

At $T = 19$, the live block-subspace calculation gives bands 0.0381364×3, 0.1133266×5, 0.2081904×3, and 0.2385063×4. The second-to-first ratio is 2.9716 and the weighted $3+4$ center ratio is 5.9133.

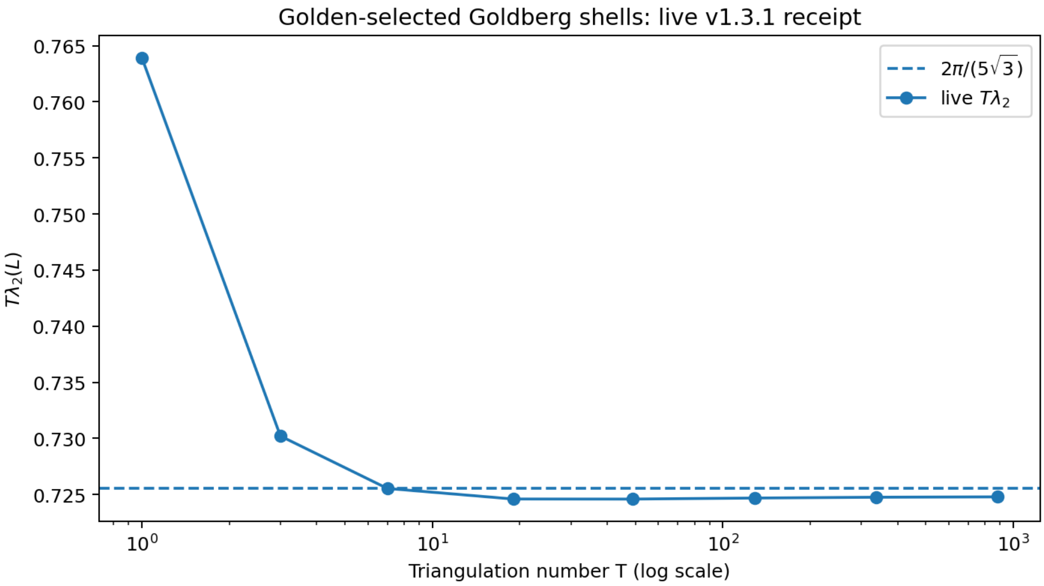

Figure 7.1: The live golden-selected shell receipt. The dashed line is the continuum matching coefficient, not an exact finite-shell target.

7.4 The honeycomb-to-sphere matching calculation

For three bond vectors $\delta_i = a\hat{e}_i$ at 120°,

$$Lf(x) = \sum_{i=1}^{3}(f(x) - f(x+\delta_i)) \tag{7.2}$$

$$= -\sum_i \delta_i \cdot \nabla f - \frac{1}{2}\sum_i(\delta_i \cdot \nabla)^2 f + \mathcal{O}(a^3). \tag{7.3}$$

Since $\sum_i \delta_i = 0$ and

$$\sum_i(\hat{e}_i \cdot \nabla)^2 = \frac{3}{2}\nabla^2, \tag{7.4}$$

we get

$$L \approx -\frac{3a^2}{4}\nabla^2. \tag{7.5}$$

The honeycomb area per vertex is $3\sqrt{3}a^2/4$. Matching $20T$ sites to a sphere gives

$$20T\frac{3\sqrt{3}}{4}a^2 = 4\pi R^2, \qquad \frac{a^2}{R^2} = \frac{4\pi}{15\sqrt{3}\,T}. \tag{7.6}$$

The $\ell = 1$ sphere eigenvalue is $2/R^2$, so the matched coefficient is

$$\boxed{T\lambda_2 \sim \frac{3a^2}{4}\frac{2}{R^2}T = \frac{2\pi}{5\sqrt{3}} = 0.7255197456937\ldots} \tag{7.7}$$

v1.3.2 correction

Every algebraic equality inside Equation (7.7) is exact once the Taylor stencil, area matching, and identification of the first graph mode with the $\ell = 1$ sphere mode are assumed. The arrow from a particular discrete fullerene family to this value is not proved by those assumptions. v1.3.2 labels the number *conditional asymptotic prediction* and the graph data *computed trend.* The Omori–Naito–Tate theorem proves low combinatorial-Laplacian eigenvalues tend to zero for specified Goldberg–Coxeter limits, but it does not by itself establish this normalized universal constant for the golden-selected family.

7.5 Operator dependence remains open

The live golden sequence approaches the low 0.7248 region, while the deeper leapfrog receipt in the core scroll reports 0.724799130 at $T = 2187$. Whether the limiting coefficient is $2\pi/(5\sqrt{3})$, another family-dependent constant, or a coefficient with slowly decaying defect corrections is an open numerical and analytic question. Required tests include:

1. run WELD, leapfrog, class-I, class-II, and golden-selected families under one eigensolver and one normalization;
2. fit corrections only after plotting residuals against several plausible scales ($T^{-1/2}$, T^{-1}, defect separation, chirality);
3. vary the discrete Laplacian (combinatorial, normalized, cotangent/geometric) and document which constant changes;
4. prove convergence before replacing “trend” by “limit.”

Chapter 8

The Shared Quadratic Form and the $SU(3)$ Boundary

8.1 What is exact

The dominant quadratic part of the $SU(3)$ quadratic Casimir for Dynkin labels (p, q) is

$$T(p, q) = p^2 + pq + q^2, \tag{8.1}$$

and

$$\boxed{C_2(p, q) = \frac{1}{3}(p^2 + pq + q^2 + 3p + 3q).} \tag{8.2}$$

The representation dimension is

$$\boxed{\dim(p, q) = \frac{1}{2}(p + 1)(q + 1)(p + q + 2).} \tag{8.3}$$

Examples are

$$(1, 0) \mapsto \mathbf{3}, \quad (1, 1) \mapsto \mathbf{8}, \quad (2, 1) \mapsto \mathbf{15}, \quad (3, 2) \mapsto \mathbf{42}. \tag{8.4}$$

8.2 Root lattice versus weight lattice

Let $Q(A_2)$ be the root lattice and $P(A_2)$ the weight lattice. They are dual up to the conventional inner-product normalization, and

$$P(A_2)/Q(A_2) \cong \mathbb{Z}/3\mathbb{Z}. \tag{8.5}$$

Thus $[P : Q] = 3$. Both are hexagonal lattices related by rotation and scaling, but they are not literally the same subset of a fixed Euclidean vector space.

v1.3.2 correction

Replace "the A_2 root lattice – which is the $SU(3)$ weight lattice" by: "the Goldberg norm is the standard hexagonal A_2 quadratic form; after a stated normalization, the same form is the quadratic part of the $SU(3)$ Casimir on Dynkin labels. The root and weight lattices are dual and commensurable, with index three."

8.3 Cross-indexing is not particle identification

The pair $(1, 1)$ has $T = 3$, so its Goldberg shell count is C_{60}. The same pair labels the adjoint $SU(3)$ representation of dimension eight. This is an exact reuse of an integer pair in two constructions. It does not imply that C_{60} is the gluon octet, that QCD lives on the fullerene shell, or that a hadron mass follows from T.

Chapter 9

Klein's Icosahedral Forms

9.1 The three binary forms

In one classical normalization,

$$
\begin{aligned}
f_{12}(z,w) &= zw(z^{10} + 11z^5w^5 - w^{10}), && (9.1)\\
H_{20}(z,w) &= -(z^{20} + w^{20}) + 228(z^{15}w^5 - z^5w^{15}) - 494z^{10}w^{10}, && (9.2)\\
T_{30}(z,w) &= z^{30} + 522z^{25}w^5 - 10005z^{20}w^{10} && (9.3)\\
&\quad - 10005z^{10}w^{20} - 522z^5w^{25} + w^{30}. && (9.4)
\end{aligned}
$$

The independent symbolic expansion gives

Exact statement

$$\boxed{T_{30}^2 + H_{20}^3 = 1728 f_{12}^5.} \tag{9.5}$$

No random sampling is required for the identity.

9.2 Why the degrees are 12, 20, and 30

The exceptional orbits of the rotational icosahedral group have sizes 12 vertices, 20 face centers, and 30 edge midpoints. The degrees of the fundamental forms encode these orbit divisors. Their equality with $P(C_{20}) = 12$, $V(C_{20}) = 20$, and $E(C_{20}) = 30$ is therefore geometric group structure, not a free-standing numerical coincidence.

9.3 The vertex rings

The affine roots of $f_{12}(z,1)$ include $z = 0$ and the roots of

$$z^{10} + 11z^5 - 1 = 0. \tag{9.6}$$

Writing $y = z^5$,

$$y_\pm = \frac{-11 \pm 5\sqrt{5}}{2}. \tag{9.7}$$

Their fifth-root radii are

$$|y_+|^{1/5} = \varphi^{-1}, \qquad |y_-|^{1/5} = \varphi. \tag{9.8}$$

Together with 0 and ∞, this produces two five-point rings plus the poles.

9.4 Gradient equivariants

For a homogeneous invariant F, define the projective map

$$g_F = [F_w : -F_z]. \tag{9.9}$$

The source's sparse degree-11, degree-19, and degree-29 polynomial pairs were compared symbolically with the gradients. They agree projectively, with scalar factors 1, 20, and 30 respectively.

9.5 Basins: what the finite experiment can say

The HTML iterates random starting points and clusters final directions. This can discover 20, 12, and 32 attractors under sufficient coverage. It cannot prove that every attractor was sampled, that a chosen clustering threshold is canonical, or that basin areas have a fixed range unless those areas are actually computed in the same run.

v1.3.2 correction

The hard-coded row "basin areas are wildly unequal, 0.12% ... 13.25%" is not produced by the current Section XIII algorithm. Either add a deterministic area estimator with a receipt or remove the row. v1.3.2 treats the orbit counts as reproducible numerical evidence and the projective-gradient formulas as exact.

Chapter 10

Numerical Architecture: Where Exactness Actually Breaks

10.1 Three distinct float walls

The browser section combined several different notions of resolution. v1.3.2 keeps them separate:
1. relative spacing near one: $\varepsilon_{64} = 2^{-52}$;
2. exact representation of all integers only through 2^{53};
3. exact representation of selected larger integers that happen to be multiples of the local spacing.
One cannot infer the exact-integer failure index solely from ε_{64}.

10.2 Exact break indices

Using exact Python integers as ground truth and IEEE binary64 conversion:

quantity	last level satisfying the stated exactness test
T_n represented exactly	$n = 38$
$V_n = 20T_n$ represented exactly	$n = 36$
$E_n = 30T_n$ represented exactly	$n = 35$
$\chi = 20T - 30T + (10T + 2)$ evaluates to 2	$n = 35$
first failed χ	$n = 36$, result 0

The composite invariant can fail before T_n because subtraction magnifies loss of low bits.

10.3 Corrected forward-error measurement

The v1.3.1 code formed

$$\frac{|T_n^{\mathrm{f64}} - \mathrm{Number}(T_n^{\mathrm{BigInt}})|}{\mathrm{Number}(T_n^{\mathrm{BigInt}})}. \tag{10.1}$$

Both operands may share the same rounded binary64 value. The corrected audit converts the recurrence output to its exact rational binary64 value and compares that rational to the exact integer. The first nonzero error occurs at $n = 39$ and the maximum through $n = 159$ is

$$1.0189 \times 10^{-15}. \tag{10.2}$$

The conclusion of forward stability survives; the original error meter was not a valid independent reference.

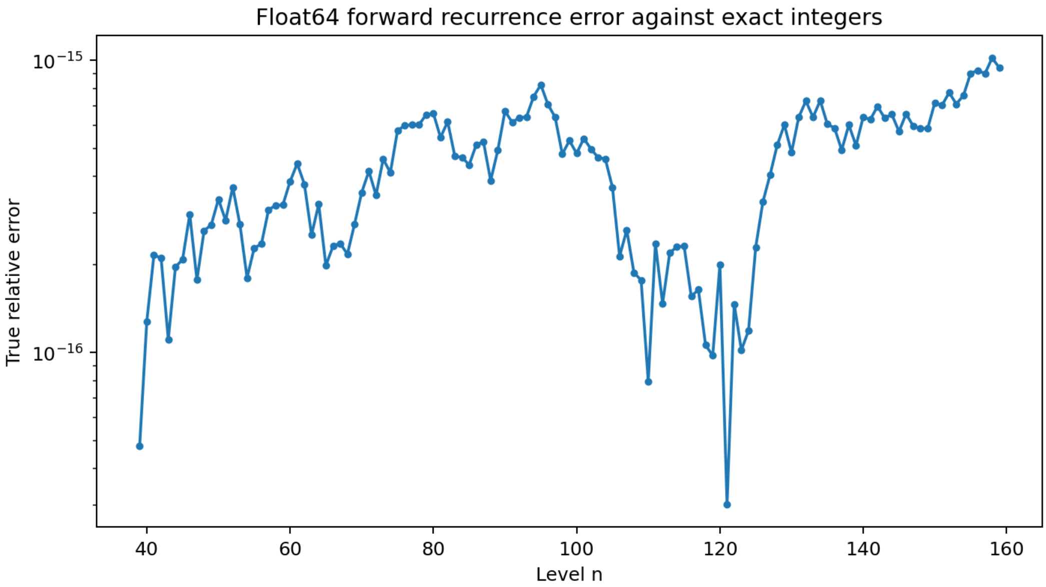

Figure 10.1: True relative error of the forward recurrence against exact integers. Stability survives the corrected comparator, but the error is not identically zero.

10.4 Backward instability

Starting from rounded values at $n = 60$ and marching

$$T_n = 2T_{n+2} + 2T_{n+1} - T_{n+3} \tag{10.3}$$

backward produces relative errors approximately

$$1.8 \times 10^{-10}\ (n = 50), \quad 9.5 \times 10^{6}\ (n = 30), \quad 5.0 \times 10^{23}\ (n = 10), \quad 1.2 \times 10^{32}\ (n = 0). \tag{10.4}$$

The same recurrence is stable in one direction and catastrophically unstable in the other because the dominant and reciprocal modes exchange roles.

10.5 Precision formats

For a format with p significand bits, the scale budget measured in φ^2 rungs is

$$N_{\text{rungs}}(p) = \frac{p \ln 2}{\ln \varphi^2}. \tag{10.5}$$

format	precision bits	rung budget	enough for 147 scale rungs?
binary16	11	7.9	no
binary32	24	17.3	no
binary64	53	38.2	no
x87 extended	64	46.1	no
binary128	113	81.4	no
binary256	237	170.7	yes

The exact integer T_{147} has 205 bits. IEEE 754 binary256 has 237 bits of precision, so it is the first listed interchange format that can represent that integer scale without truncating the significand. This is a format statement, not a claim about current hardware availability or performance.

10.6 The Planck-to-horizon ruler

With the chosen values $\ell_P = 1.616255 \times 10^{-35}\,\mathrm{m}$ and $L_H = 4.4 \times 10^{26}\,\mathrm{m}$,

$$\frac{\ln(L_H/\ell_P)}{\ln\varphi^2} = 146.9822\ldots \tag{10.6}$$

This is a dimensionless scale-counting exercise. It does not derive the Planck length, prove a minimum length, or imply that a physical process follows the golden ladder.

10.7 Selected precision benchmarks, updated to 9 August 2026

The v1.3.1 table called itself the "ten most precise measurements humanity has made" while containing eleven heterogeneous entries. v1.3.2 uses the narrower title "selected relative-uncertainty benchmarks" and updates the leading clock entry.

benchmark	relative uncertainty	$N(u)$ rungs	source
Cryogenic $^{40}Ca^+$ ion clock	4.40×10^{-19}	43.92	Zhang et al., PRL 136 (2026) 053202
Multi-ion Sr^+ clock	5.30×10^{-19}	43.72	Filzinger et al., arXiv:2603.23446
$^{27}Al^+$ single-ion clock	5.50×10^{-19}	43.69	Marshall et al., PRL 2025
$^{88}Sr^+$ single-ion clock	7.90×10^{-19}	43.31	Lindvall et al., 2025
^{87}Sr lattice clock	8.10×10^{-19}	43.28	Aeppli et al., PRL 133 (2024) 023401
In^+/Yb^+ crystal clock	2.50×10^{-18}	42.11	Hausser et al., PRL 134 (2025) 023201
$^{88}Sr^+$ absolute frequency vs TAI	9.80×10^{-17}	38.30	Lindvall et al., 2025
Hydrogen $1S$-$2S$	1.40×10^{-14}	33.15	Parthey et al., PRL 107 (2011) 203001
Electron magnetic moment $g/2$	1.30×10^{-13}	30.83	Fan et al., PRL 130 (2023) 071801
Muon anomaly final precision	1.27×10^{-7}	16.50	Fermilab Muon g-2 final result
Newtonian constant G	2.20×10^{-5}	11.14	CODATA 2022

The table compares published uncertainties under one logarithmic mapping. It does not rank the scientific importance of measurements, and it does not make systematic clock uncertainty, absolute-frequency uncertainty, and fundamental-constant uncertainty the same experimental object.

Chapter 11

The LUCA Spiral Audit and the Handoff to v1.3.2

11.1 The EML operator

The external conjecture defines

$$\boxed{\mathrm{eml}(x,y) = e^x - \operatorname{Log} y}, \qquad S \to 1 \mid \mathrm{eml}(S,S). \tag{11.1}$$

The immediate identities are

$$e = \mathrm{eml}(1,1), \qquad e^x = \mathrm{eml}(x,1). \tag{11.2}$$

For positive real x,

$$\operatorname{Log} x = \mathrm{eml}\big(1, \mathrm{eml}(\mathrm{eml}(1,x),1)\big) \tag{11.3}$$

$$= e - \operatorname{Log}(e^e/x) = \log x. \tag{11.4}$$

11.2 Domain and branch ledger

Over $\mathbb{C}$, $\exp(\operatorname{Log} z) = z$ for nonzero z on the principal logarithm, but $\operatorname{Log}(e^z) = z$ only inside a principal strip, modulo branch jumps. Consequently:

- the log chain is exact on stated domains/sheets, not a single global identity of principal branches;
- subtraction $x - y = \mathrm{eml}(\operatorname{Log} x, e^y)$ is exact for the browser's real test domain, while complex y requires strip control;
- the code's convention $\operatorname{Log} 0 = -\infty$ and $e^{-\infty} = 0$ is an extended-real computational lane, not ordinary complex analyticity at zero;
- constructing i from $(-1)^{1/2}$ is branch-dependent; branch-robust downstream real formulas must be checked under conjugation.

v1.3.2 correction

Replace the blanket sentence "the identities on every node are classical algebra over $\mathbb{C}$ (principal branch)" by a per-node domain contract. Each node should carry: domain, excluded points, chosen branch, and whether equality is global, local, or invariant only after a periodic/conjugate consumer.

11.3 What the browser verification proves

LUCA v0.1/v0.2 evaluates chains at 16 deterministic sample points and reports

$$D = -\log_{10}(\text{relative error}), \qquad D \le 15.9. \tag{11.5}$$

A high D is a finite numerical certificate at the sampled points. It is not a symbolic proof of functional identity. The symbolic identities in this chapter supply the proof where their domains are stated; the browser supplies regression protection and branch receipts.

11.4 The light-matrix content transferred from LUCA v0.2

The LUCA golden lane correctly contributes four items:

1. Faddeev–LeVerrier construction of the coefficient list $(1, -3, 0, 3, -1)$;
2. reciprocal-spectrum explanation from anti-palindromy;
3. the exact angle $\theta_\varphi = \arctan(\sqrt{15} - 2\sqrt{3})$;
4. a target/current/error display for the ratio and angular zigzag.

These are integrated into Chapters 4 and 5.

11.5 Finite-N Vogel experiment

For $N = 33$ points placed by

$$r_k = \sqrt{k}, \qquad \theta_k = k\alpha, \tag{11.6}$$

the live v0.2 sweep maximizes minimum pairwise separation near

$$\alpha_{33} = 137.6066065^\circ, \tag{11.7}$$

not at the golden angle

$$\alpha_\varphi = \frac{360^\circ}{\varphi^2} = 137.5077641^\circ. \tag{11.8}$$

The difference is about 0.09884°. This refutes an unqualified finite-N optimum claim. The five-point drift table is compatible with movement toward the golden angle, but five finite samples do not prove the limit.

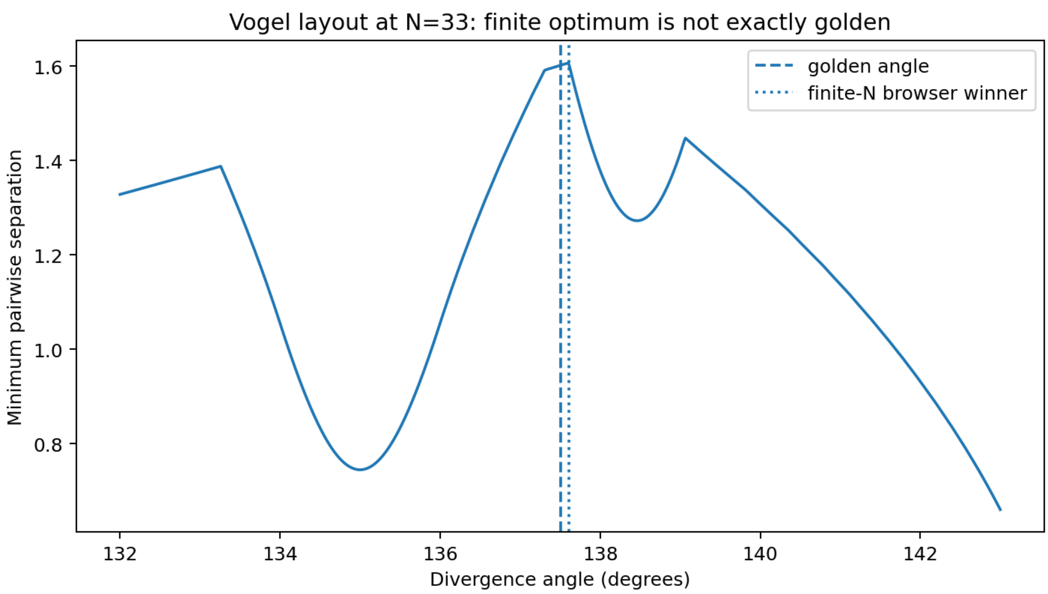

Figure 11.1: The finite-N objective is jagged and its maximizer is not exactly the golden angle.

11.6 Leaf-count null and the 32/33 discrepancy

Using all 33 live spiral nodes, the independent reconstruction of the v0.2 leaf recurrences gives

$$\text{growth} = 1.159672, \quad 95\% \text{ descriptive interval } [1.140910, 1.178742], \quad R^2 = 0.9109. \tag{11.9}$$

Excluding the first constant e gives the 32-point values printed in the handoff scroll:

$$\text{growth} = 1.155084, \quad [1.135961, 1.174529], \quad R^2 = 0.9052. \tag{11.10}$$

Both exclude φ by roughly forty fitted standard errors. The null result is robust; the sample definition was not.

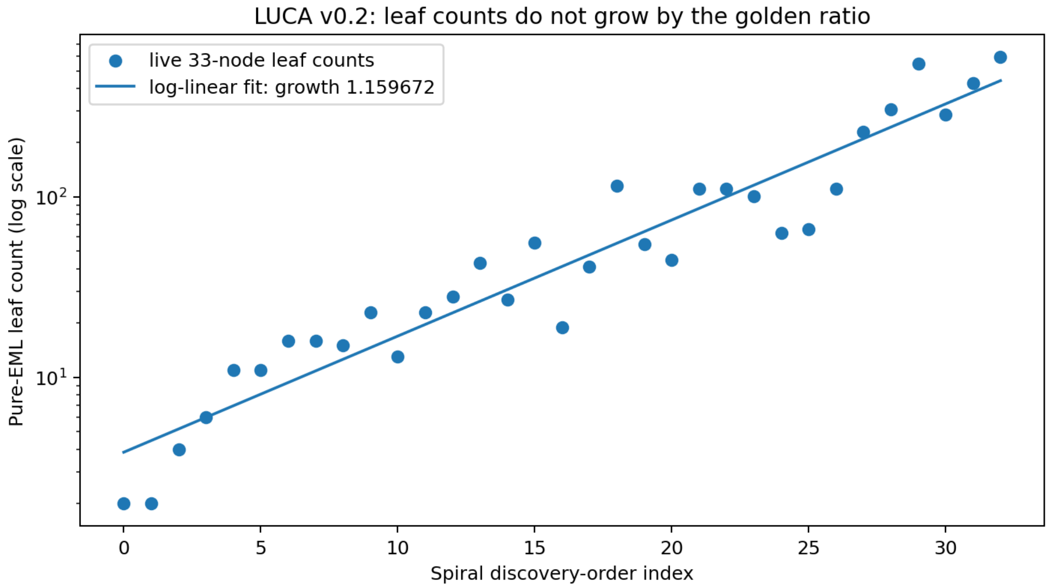

Figure 11.2: The live 33-node leaf counts and their descriptive log-linear fit. Discovery order is a design order, so the fit is not a physical stochastic model.

Chapter 12

The v1.3.2 Integration Contract

12.1 Section map

The recommended living-paper order is:

1. Birth / abstract and status grammar.
2. Constants φ and π.
3. Euler and discrete Gauss–Bonnet.
4. Hexagonal closure norm, matrix, composition, and angle.
5. Light matrix built by Faddeev–LeVerrier.
6. Golden selector, shell catalogue, and selected-not-nested theorem.
7. Golden-ray angle and signed convergence plot.
8. Exact C_{60} adjacency and Laplacian certificate.
9. Graph-tower lane selector: golden, leapfrog, WELD, fixed GC.
10. Conditional continuum matching and finite receipts.
11. $A_2/SU(3)$ quadratic-form correspondence with corrected lattice language.
12. Klein forms and exact symbolic syzygy.
13. Equivariant numerical experiment with deterministic seed/coverage ledger.
14. Float architecture with corrected exact reference.
15. Selected precision benchmarks, dated.
16. LUCA/EML handoff and branch ledger.
17. Honest physical boundary.
18. Reproducibility manifest and frozen lineage.

12.2 HUD contract

Every live shell should show independent rows:

```
CLOSURE pair (k,l), T, V/E/F/P/H, chi, boundary, degree, cert hash
GOLDEN target phi, current k/l, signed error, tolerance, lock state
ANGLE target theta_phi, current theta(k,l), signed error,
         error ratio -> -phi^-2, suppress below numerical floor
SPECTRUM family name, lambda2, T*lambda2, previous delta,
         low-band multiplicities, solver tolerance
BOUNDARY EXACT / COMPUTED / DESIGN / HYPOTHESIS / EXTERNAL
```

No lock may be true while the displayed error exceeds its tolerance. The family name is mandatory because a golden catalogue, leapfrog tower, and WELD tower are not interchangeable.

12.3 Compute gate

Before allocating a shell, predict

$$V_{\text{next}} = SV_{\text{current}} \tag{12.1}$$

for a fixed area multiplier S, or compute $20T(k', \ell')$ for a selected pair. Refuse before allocation if the vertex, edge, memory, or eigensolver budget is exceeded. The refusal must print the predicted price.

12.4 Deterministic receipt schema


```
{
  "schema": "thea.light-matrix.receipt.v1.3.2",
  "family": "golden-selected | leapfrog | weld | gc-fixed",
  "pair": {"k": "21", "ell": "13"},
  "counts": {"T": "883", "V": "17660", "E": "26490",
             "F": "8832", "P": "12", "H": "8820", "chi": "2"},
  "topology": {"boundary_edges": 0, "nonmanifold_edges": 0,
               "degree3_vertices": 17660, "connected": true},
  "golden": {"target": 1.6180339887498948,
             "current": 1.6153846153846154,
             "signed_error": -0.0026493733652794},
  "angle": {"target_deg": 22.23875609296496,
            "current_deg": 22.264023, "signed_error_deg": 0.025267},
  "spectrum": {"method": "Lanczos full reorthogonalization",
               "lambda2": 0.0008208191343368484,
               "T_lambda2": 0.7247832956194371,
               "residual_tolerance": 1e-10},
  "status": {"closure": "EXACT", "spectrum": "COMPUTED"}
}
```


The timestamp belongs outside the hashed mathematical payload. Random experiments must carry a fixed seed or deterministic point set.

12.5 Regression tests

At minimum:

1. exact integer tests for norm, metric similarity, composition, and shell counts;
2. exact characteristic polynomial of $\mathcal{M}_{\text{light}}$ and coefficient anti-palindromy;
3. exact golden-angle radical equality;
4. exact C_{60} graph counts and characteristic factorization;
5. exact Klein syzygy and projective-gradient checks;
6. branch-domain tests for EML nodes, including excluded points;
7. deterministic spectral smoke tests for C_{20}, C_{60}, C_{140};
8. float comparator test against rationalized binary64, not rounded exact operands;
9. byte scan for malformed Unicode and line endings;
10. certificate regeneration on a second path with identical math hash.

12.6 Honest physical boundary

Honest boundary

This tower does not derive a particle mass, coupling constant, cross-section, decay rate, Planck length, or Standard Model observable. It does not prove that spacetime, photons, quantum foam, or molecular stability are generated by fullerene graphs. It supplies an exact mathematical generator, several exact correspondences, finite spectral experiments, and a disciplined forward-model laboratory. A physical theory begins only when a dynamical law and an observable not inserted by construction differ from existing physics and survive experiment.

Chapter 13

Copy-Ready Formula Tower

13.1 Core constants and topology

$$\varphi = \frac{1+\sqrt{5}}{2} = 2\cos\frac{\pi}{5}, \qquad \varphi^2 = \varphi + 1, \qquad \varphi^{-2} = 2 - \varphi, \tag{13.1}$$

$$3V = 2E, \qquad 5P + 6H = 2E, \qquad V - E + P + H = 2, \tag{13.2}$$

$$P = 12, \qquad E = \frac{3V}{2}, \qquad H = \frac{V}{2} - 10, \tag{13.3}$$

$$\frac{\pi}{3}\sum_f (6 - p_f) = 4\pi. \tag{13.4}$$

13.2 Closure ring

$$\zeta_6 = e^{i\pi/3}, \qquad T = k^2 + k\ell + \ell^2, \tag{13.5}$$

$$M_{k,\ell} = \begin{pmatrix} k & -\ell \\ \ell & k+\ell \end{pmatrix}, \qquad \det M_{k,\ell} = T, \tag{13.6}$$

$$M_{k,\ell}^{\mathsf{T}} Q M_{k,\ell} = TQ, \qquad Q = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix}, \tag{13.7}$$

$$(k', \ell') = (ak - b\ell, a\ell + bk + b\ell), \qquad T' = (a^2 + ab + b^2)T, \tag{13.8}$$

$$\theta_{k,\ell} = \operatorname{atan2}(\sqrt{3}\ell, 2k + \ell). \tag{13.9}$$

13.3 Golden lane

$$\begin{pmatrix} k_{n+1} \\ \ell_{n+1} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} k_n \\ \ell_n \end{pmatrix}, \qquad (k_n, \ell_n) = (F_{n+1}, F_n), \tag{13.10}$$

$$T_n = k_n^2 + k_n \ell_n + \ell_n^2, \qquad T_{n+3} = 2T_{n+2} + 2T_{n+1} - T_n, \tag{13.11}$$

$$T_n = \frac{2}{5}(\varphi^{2n+2} + \varphi^{-2n-2}) - \frac{1}{5}(-1)^n, \qquad \frac{T_{n+1}}{T_n} \to \varphi^2, \tag{13.12}$$

$$\theta_\varphi = \arctan(\sqrt{15} - 2\sqrt{3}), \qquad \frac{\theta_{n+1} - \theta_\varphi}{\theta_n - \theta_\varphi} \to -\varphi^{-2}. \tag{13.13}$$

13.4 Light matrix

$$\mathcal{M}_{\text{light}} = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{13.14}$$

$$\det(\lambda I - \mathcal{M}_{\text{light}}) = (\lambda - 1)(\lambda + 1)(\lambda^2 - 3\lambda + 1), \tag{13.15}$$

$$\operatorname{spec}(\mathcal{M}_{\text{light}}) = \{\varphi^2, 1, -1, \varphi^{-2}\}, \tag{13.16}$$

$$p(\lambda) = \lambda^4 - 3\lambda^3 + 3\lambda - 1 = -\lambda^4 p(1/\lambda). \tag{13.17}$$

13.5 C_{60} and graph spectra

$$L = 3I - A, \tag{13.18}$$

$$\lambda_{\min}(A_{60}) = -\varphi^2, \tag{13.19}$$

$$\lambda_2(L_{60}) = \frac{9}{4} - \frac{\sqrt{2}}{8}\left(\sqrt{10} + 2\sqrt{19 - \sqrt{5}}\right), \tag{13.20}$$

$$T\lambda_2 \sim \frac{2\pi}{5\sqrt{3}} \quad \text{under the stated continuum matching assumptions.} \tag{13.21}$$

13.6 A_2, $SU(3)$, and Klein

$$T(p, q) = p^2 + pq + q^2, \tag{13.22}$$

$$C_2(p, q) = \frac{1}{3}\left(T(p, q) + 3p + 3q\right), \tag{13.23}$$

$$\dim(p, q) = \frac{1}{2}(p + 1)(q + 1)(p + q + 2), \tag{13.24}$$

$$T_{30}^2 + H_{20}^3 = 1728 f_{12}^5, \tag{13.25}$$

$$g_F = [F_w : -F_z]. \tag{13.26}$$

Chapter 14

Independent Audit Receipt

The following table is generated from `light_matrix_v1.3.2_audit_receipt.json`. Long result fields are truncated here; the JSON preserves the full values.

#	check	status	verdict	result
1	Euler forces twelve pentagons	EXACT	PASS	{'forced_pentagons': 12, 'solution': {'E': '3*H + 30', 'P': '12', 'V': '2*H + 20'}}
2	C60 combinatorial counts from T=3	EXACT	PASS	{'E': 90, 'F': 32, 'H': 20, 'P': 12, 'T': 3, 'V': 60, 'chi': 2}
3	discrete Gauss-Bonnet charge	EXACT	PASS	4*pi
4	hexagonal closure determinant	EXACT	PASS	det M(k,l) = k^2 + k l + l^2
5	hexagonal metric similarity	EXACT	PASS	M^T Q M = T Q
6	closure matrix composition	EXACT	PASS	{'ell_prime': 'a*l + b*k + b*l', 'k_prime': 'a*k - b*l'}
7	hexagonal norm multiplicativity	EXACT	PASS	N(gz)=N(g)N(z)
8	leapfrog multiplier and count tower	EXACT	PASS	{'area_multiplier': 3, 'linear_multiplier': 1.7320508075688772, 'vertex_sequence': [20, 60, 180, 540, 1620]}
9	selected is not fixed golden nesting	EXACT / CORRECTION	PASS	{'argument': 'fixed GC area multiplier is an integer S; sqrt(S)=phi would require S=phi^2 irrational'}
10	Fibonacci selector pairs	EXACT	PASS	[[1, 0], [1, 1], [2, 1], [3, 2], [5, 3], [8, 5], [13, 8], [21, 13]]
11	projective golden error identity	EXACT	PASS	r_{n+1}-phi = -(r_n-phi)/(phi r_n)
12	golden triangulation sequence	EXACT	PASS	[1, 3, 7, 19, 49, 129, 337, 883, 2311, 6051]
13	golden T recurrence	EXACT	PASS	T[n+3] = 2T[n+2] + 2T[n+1] - T[n]
14	golden T closed form	EXACT	PASS	{'last': 91530451, 'verified_levels': 20}
15	T ratio alternates rather than climbs monotonically	EXACT / CORRECTION	PASS	{'first_ratios': [2.3333333333333335, 2.7142857142857144, 2.5789473684210527, 2.63265306122449, 2.612403100775194, 2.620178041543027], 'l...
16	golden-ray radical angle	EXACT	PASS	{'degrees': 22.23875609296496, 'difference_between_forms': 3.3306690738754696e-16, 'radians': 0.3881395153701887}
17	golden-ray angular contraction	EXACT + COMPUTED	PASS	{'last_reported_ratio': -0.3819660304017546, 'levels': [3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18], 'target': -0.3819660112...
18	symmetric-square light-matrix lift	EXACT	PASS	s(k+l,k,P)=M_light s(k,l,P)
19	light-matrix Faddeev-LeVerrier coefficient list	EXACT	PASS	[1, -3, 0, 3, -1]
20	light-matrix characteristic factorization	EXACT	PASS	(lambda - 1)*(lambda + 1)*(lambda**2 - 3*lambda + 1)
21	light-matrix anti-palindromy	EXACT	PASS	p(lambda) = -lambda^4 p(1/lambda)
22	light-matrix reciprocal spectrum	EXACT	PASS	['sqrt(5)/2 + 3/2', '1', '-1', '3/2 - sqrt(5)/2']
23	light-matrix eigenbasis	EXACT	PASS	{'-1': True, '1': True, 'phi^-2': True, 'phi^2': True}
24	C60 graph topology certificate	EXACT	PASS	{'E': 90, 'F': 32, 'V': 60, 'chi': 2, 'connected': True, 'degree_set': [3], 'face_counts': {'5': 12, '6': 20}}
25	C60 complete adjacency characteristic polynomial	EXACT	PASS	(lambda - 3)*(lambda - 1)**9*(lambda + 2)**4*(lambda**2 - lambda - 3)**5*(lambda**2 + lambda - 4)**4*(lambda**2 + lambda - 1)**5*(lambda*...
26	C60 minimum adjacency eigenvalue	EXACT + COMPUTED	PASS	{'exact': '-(3+sqrt(5))/2 = -phi^2', 'lambda_min': -2.618033988749895, 'multiplicity': 3}
27	C60 Fiedler radical and quartic	EXACT + COMPUTED	PASS	{'difference': 7.882583474838611e-15, 'numeric': 0.2434017461399247, 'quartic_residual': 8.926193117986259e-14, 'radical': 0.243401746139...
28	C60 low-band multiplicities	COMPUTED	PASS	[{'multiplicity': 3, 'value': 0.2434017461399247}, {'multiplicity': 5, 'value': 0.697224362268004}, {'multiplicity': 3, 'value': 1.179750...
29	continuum coefficient 2pi/(5sqrt3)	EXACT CONDITIONAL / CORRECTION	PASS	{'coefficient': '2*sqrt(3)*pi/15', 'numeric': 0.7255197456936872, 'status': 'conditional asymptotic prediction, not a proved discrete gra...

#	check	status	verdict	result
30	SU3 Casimir and dimension formulas	EXACT	PASS	{'C2': 'p**2/3 + p*q/3 + p + q**2/3 + q', 'dimension': '(p/2 + 1/2)*(q + 1)*(p + q + 2)', 'table': {'(0, 1)': 3, '(1, 0)': 3, '(1, 1)': 8...
31	A2 root/weight lattice index-three relation	EXACT / CORRECTION	PASS	{'Cartan_matrix': [['2', '-1'], ['-1', '2']], 'correction': 'dual/commensurable hexagonal lattices, not literally identical', 'determinan...
32	Klein icosahedral syzygy	EXACT	PASS	T_30ˆ2 + H_20ˆ3 = 1728 f_12ˆ5 (symbolically zero residual)
33	Klein gradient equivariants	EXACT	PASS	{'g11': {'degree': 11, 'projective_gradient': True, 'scale': '1'}, 'g19': {'degree': 19, 'projective_gradient': True, 'scale': '1/20'}, '...
34	float64 exactness break indices	COMPUTED / CORRECTION	PASS	{'E_last_exact': 35, 'T_last_exact': 38, 'V_last_exact': 36, 'chi_first_bad_level': 36, 'chi_first_bad_value': 0.0, 'chi_last_correct': 35}
35	forward stability and backward instability	COMPUTED / CORRECTION	PASS	{'backward_relative_errors': {'0': 1.1864584222679098e+32, '10': 4.9512758706203005e+23, '30': 9455846.679125344, '50': 1.807821865712679...
36	binary precision and T147 bit budget	EXACT + COMPUTED / CORRECTION	PASS	{'T147_bit_length': 205, 'correction': 'binary256 has 237 significand bits, not a 256-bit mantissa', 'format_precision_bits': {'binary128...
37	EML positive-real identity and branch ledger	EXACT CONDITIONAL / CORRECTION	PASS	{'Log(exp(z))-z': '-6.283185307179586j', 'boundary': 'node identities require domain and branch labels; 16-point locks are numerical siev...
38	LUCA leaf regression, Vogel sweep, and Fibonacci fingerprint	COMPUTED / REFUTED / CORRECTION	PASS	{'finite_N_verdict': 'golden angle is not the exact N=33 grid optimum', 'golden_nearest_index_gaps': {'13': 11, '3': 2, '5': 4, '8': 16},...

Chapter 15

Reproduction Protocol

15.1 Commands

```
python verify_light_matrix_v132.py
node verify_light_matrix_v131_live.js > light_matrix_v1.3.1_live_kernel_receipt.json
pdflatex THEA_LIGHT_MATRIX_v1.3.2_LATEX_TOWER.tex
pdflatex THEA_LIGHT_MATRIX_v1.3.2_LATEX_TOWER.tex
```

15.2 Version discipline

The new document does not overwrite any input. The source HTML and LUCA versions remain immutable. v1.3.2 is a derived audit artifact whose corrections are explicit, whose receipts are machine-readable, and whose build timestamp is not included in the mathematical hash.

15.3 Closing line

The pentagons hold. The hexes pay.
The ray now has a bearing.
The limit still owes its proof.

$$P = 12. \quad \chi = 2. \quad \theta_\varphi = \arctan(\sqrt{15} - 2\sqrt{3}).$$

Bibliography

[1] *THEA v3.0: The Math Core Scroll – the Light Matrix*, source file `Thea.md`, 2026.

[2] M. Goldberg, "A class of multi-symmetric polyhedra," *Tohoku Mathematical Journal*, 1937.

[3] D. L. D. Caspar and A. Klug, "Physical principles in the construction of regular viruses," *Cold Spring Harbor Symposia on Quantitative Biology* 27 (1962).

[4] P. W. Fowler, P. Hansen, and D. Stevanović, "A Note on the Smallest Eigenvalue of Fullerenes," *MATCH Communications in Mathematical and in Computer Chemistry* 48 (2003), 37–48.

[5] T. Omori, H. Naito, and T. Tate, "Eigenvalues of the Laplacian on the Goldberg–Coxeter Constructions for 3- and 4-Valent Graphs," *Electronic Journal of Combinatorics* 26(3) (2019), P3.7; arXiv:1807.10891.

[6] S. Li, "Transformation, Identification, and Inversion of Goldberg–Coxeter Fullerenes," arXiv:2303.07890.

[7] F. Klein, *Lectures on the Icosahedron and the Solution of Equations of the Fifth Degree*, 1884/1913 English edition.

[8] A. Odrzywolek, "All elementary functions from a single binary operator," arXiv:2603.21852v2, 2026.

[9] IEEE, *IEEE Standard for Floating-Point Arithmetic*, IEEE 754-2019.

[10] CODATA 2022 recommended values of the fundamental physical constants.

[11] B. Zhang et al., "Liquid-Nitrogen-Cooled $^{40}\mathrm{Ca}^+$ Ion Optical Clock with a Systematic Uncertainty of 4.4×10^{-19}," *Physical Review Letters* 136 (2026), 053202; arXiv:2506.17423.

[12] M. Filzinger et al., "A multi-ion optical clock with 5×10^{-19} uncertainty," arXiv:2603.23446, 2026.

[13] M. Marshall et al., "High-Stability Single-Ion Clock with 5.5×10^{-19} Systematic Uncertainty," *Physical Review Letters*, 2025.

[14] T. Lindvall et al., "$^{88}\mathrm{Sr}^+$ optical clock with 7.9×10^{-19} systematic uncertainty and measurement of its absolute frequency," 2025.

[15] A. Aeppli et al., "Clock with 8×10^{-19} Systematic Uncertainty," *Physical Review Letters* 133 (2024), 023401.